Network Science

Lecture 01: Introduction

Prof. Dr. Michael Granitzer

Dr. Kanishika Ghosh Dastidar

Dr. Jelena Mitrovic

Speaker notes

This introductory lecture establishes the core mental model for the course: representing the world through the lens of relational data. You should focus on understanding that a network is not just a static set of entities, but a dynamic system where the pattern of connections (the structure) determines outcomes such as information flow, resilience, and influence. By the end of this module, you will have practiced interpreting graphical data and identifying how network structure impacts observability and control.

Welcome to Network Science

> Modelling, analysing and infering digitial and real-world phenomena through networks.

Today

- what is a network?
- why do networks matter?
- what can we analyse with networks?
- what will you learn in this course?

≡

Networks

> **Guiding Question:** What do you think a network is?

Your Everyday Networks

Think about networks you encounter every day.

Speaker notes

Networks are ubiquitous in daily life, ranging from social interactions and digital information flow to the physical transport of goods and financial transactions. Categorizing these into families like social, information, economic, or infrastructure networks provides a high-level framework for understanding how different relational systems operate under similar abstract principles while serving diverse functional roles.

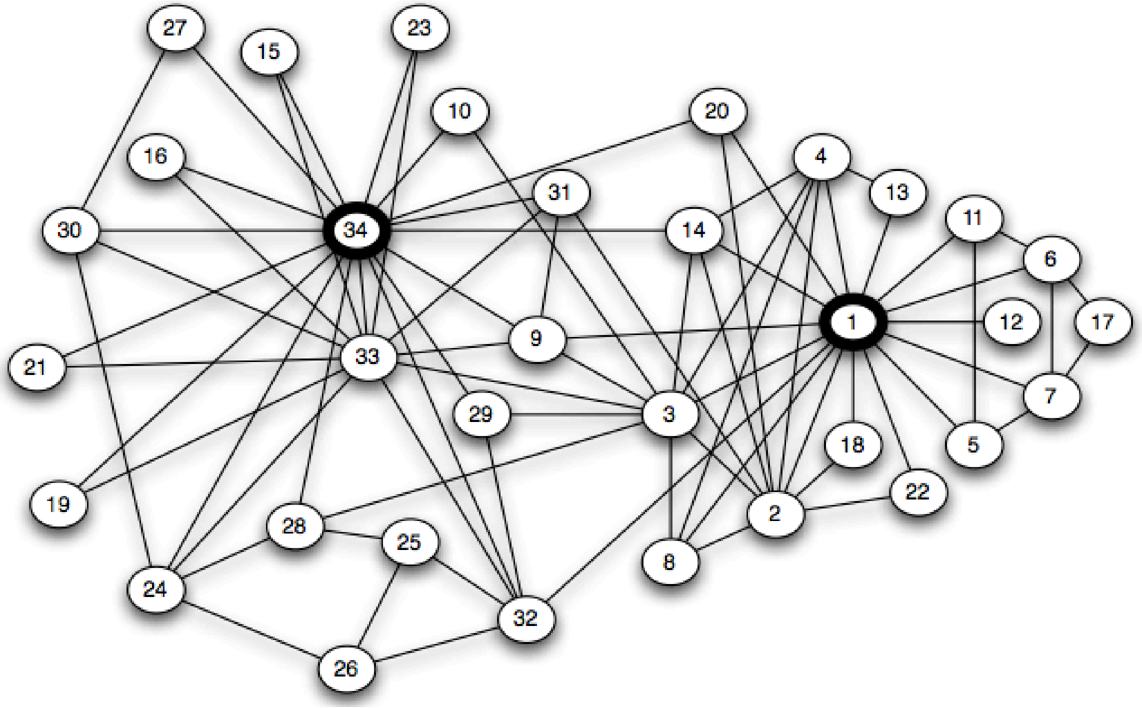
Your Everyday Networks

Think about networks you encounter every day.

- Who did you talk to today? → **social network**
- Which websites link to each other? → **information network**
- Which companies trade with each other? → **economic network**
- Which roads connect which cities? → **infrastructure network**

Social Networks

Link structures can reveal friendships, communities, and influence.



Source: Zachary 1977 — modified; Original data: Zachary 1977; visualization redrawn

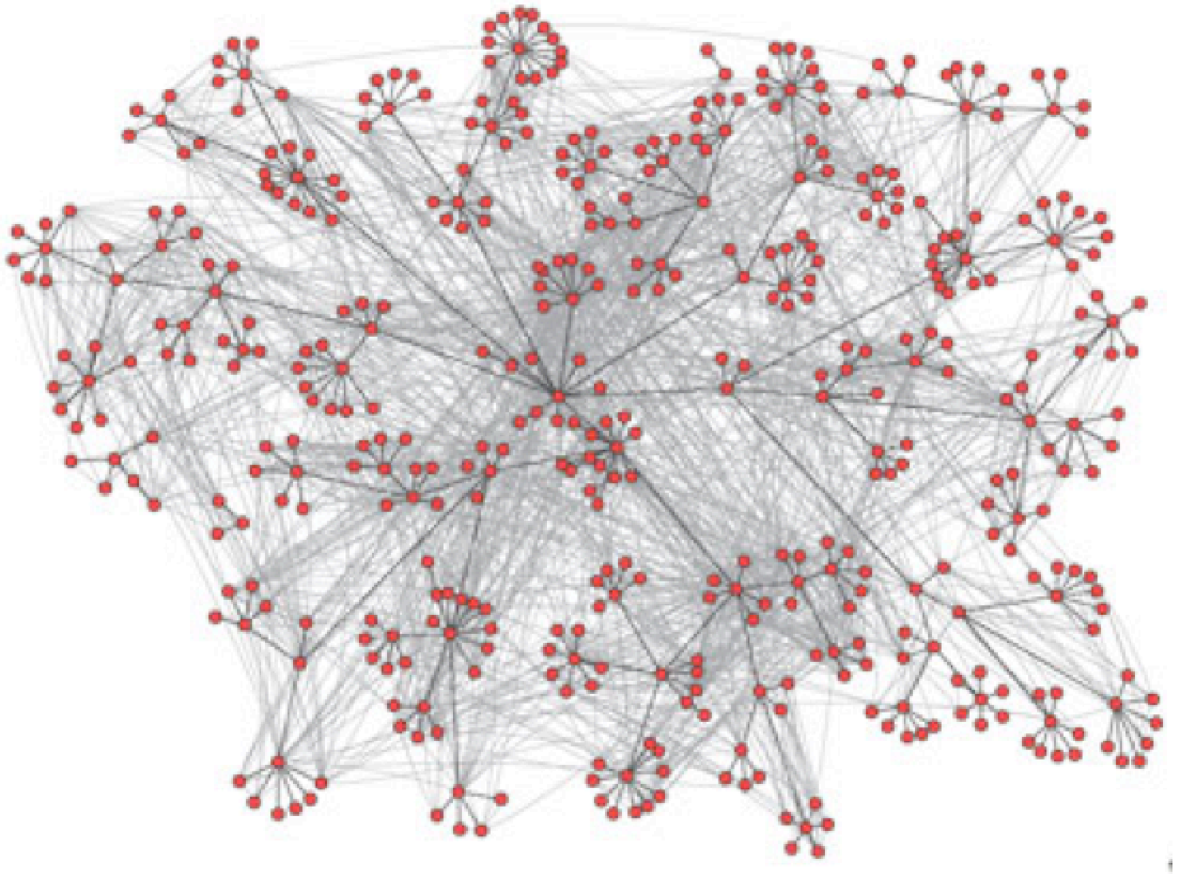
Zachary's Karate Club (1977). Each node is a club member, each edge a friendship. The club split into two factions — visible as two dense clusters. One of the most-studied small social networks.

Speaker notes

Zachary's Karate Club is a foundational case study in social network analysis. It demonstrates how simple friendship ties (edges) between individuals (nodes) can provide deep insights into group dynamics. In this historical example, the network structure correctly predicted how the club would split into two distinct factions during a leadership dispute, illustrating that relational patterns often precede and determine behavioral outcomes.

Communication Networks

Link structures can reveal information flow and potential dissemination routes.



Source: Easley & Kleinberg 2010, p. Ch. 2

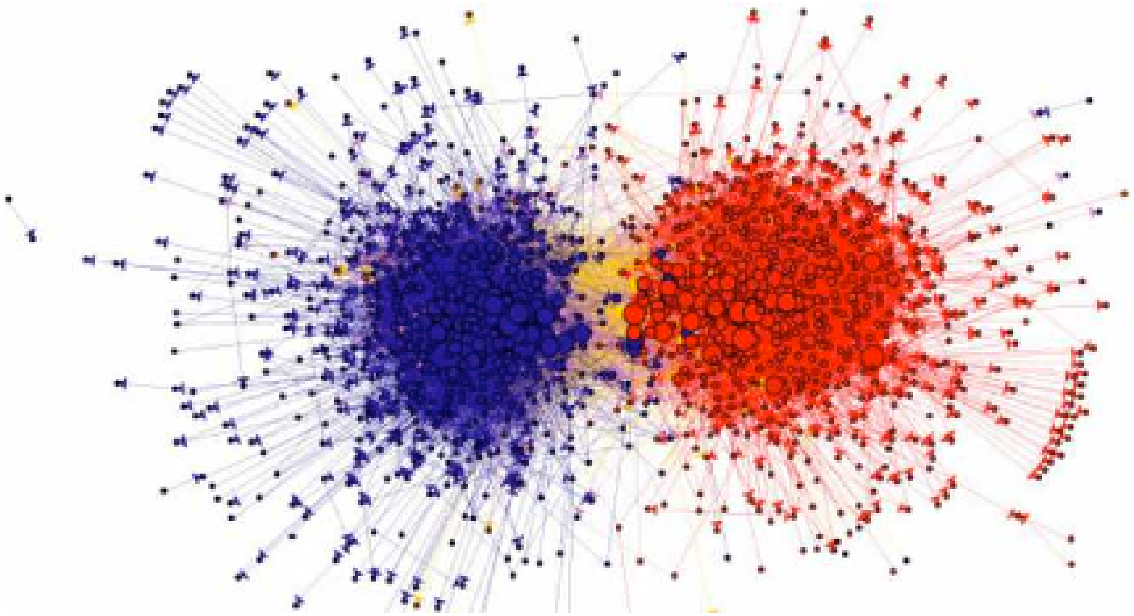
> **Email communication in an organization.** Nodes represent employees, edges represent email exchanges. The actual communication topology differs substantially from the formal hierarchy.

Speaker notes

Communication networks often reveal the "hidden" architecture of an organization. While a formal hierarchy defines the official chains of command, the actual flow of emails or information (the communication topology) often reveals a very different landscape of centers of influence and informal collaboration hubs. Identifying these discrepancies is key to understanding organizational efficiency and resilience. ::: :::

Information Networks

Link structures can reveal clusters, polarization, and community structure — before reading a single word.



Source: Adamic & Glance 2005

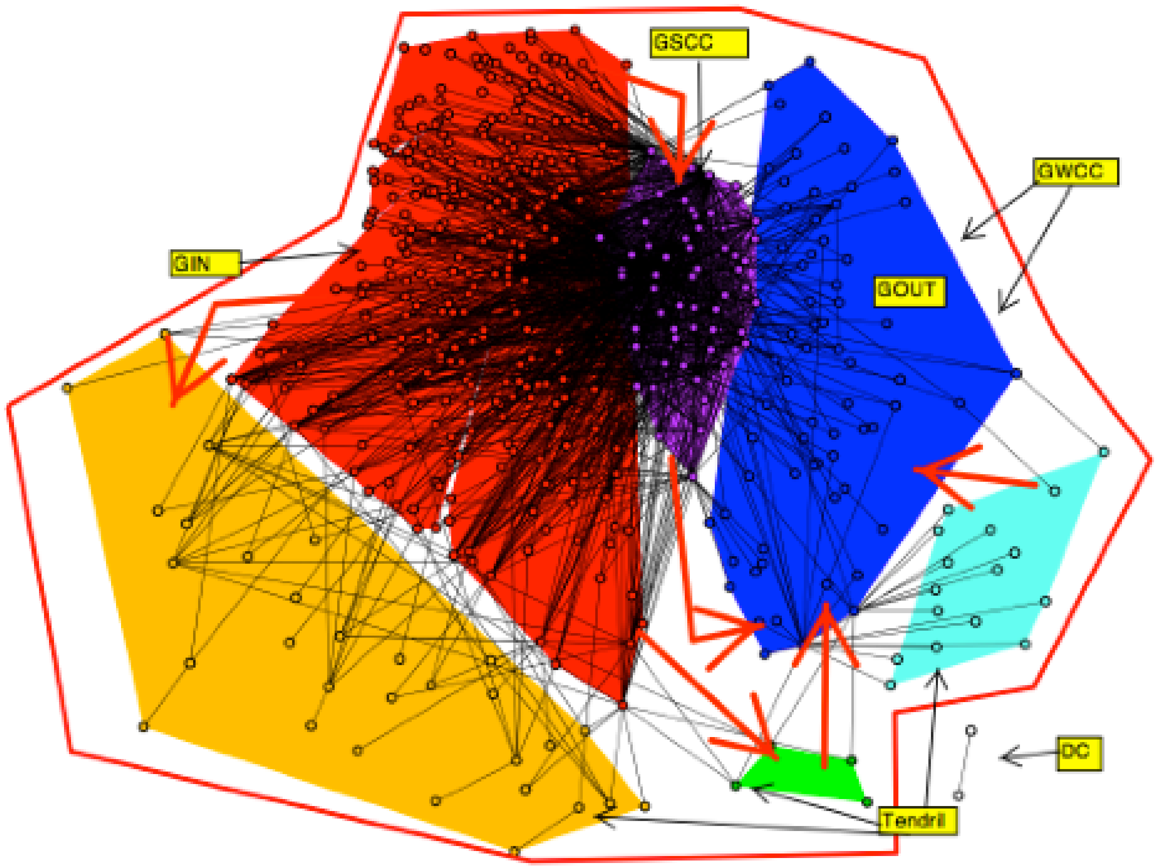
Political blog network (2004 US election). Nodes are political blogs, edges are hyperlinks between them. The two dense clusters correspond to liberal and conservative blogs — structure reveals political alignment without reading any content.

Speaker notes

Large-scale information networks, such as hyperlinks between political blogs, demonstrate how relational structure can reveal high-level properties like political polarization or community alignment. By analyzing who links to whom, we can identify "echo chambers" or clusters of influence without needing to process the underlying textual content of the pages themselves.

Economic Networks

Financial systems form networks of dependencies and influence (Easley & Kleinberg, 2010).



Source: Easley & Kleinberg 2010, p. Ch. 1

Interbank loan network. Nodes are financial institutions, edges represent lending relationships. When one institution defaults, the impact cascades through the network — structure determines systemic risk.

Speaker notes

Economic networks help visualize systemic risk. In an interbank loan network, a default at one node (bank) can trigger a cascade of failures through the edges (lending relationships). Understanding the topology of these dependencies is critical for assessing the stability of the entire financial system beyond the health of individual institutions.

What Do All These Examples Have in Common?

Speaker notes

A common thread across all domains is the formalization of relationships. Whether we are discussing people, web pages, or banks, the core abstraction remains the same: a network is a set of entities (nodes) connected by relationships (edges). This structural pattern is what allows us to apply the same mathematical tools across wildly different physical and social systems.

What Do All These Examples Have in Common?

- There are **things** (people, pages, institutions, cities).
- There are **connections** between them (friendships, links, loans, roads).
- The pattern of connections creates **structure** that matters.

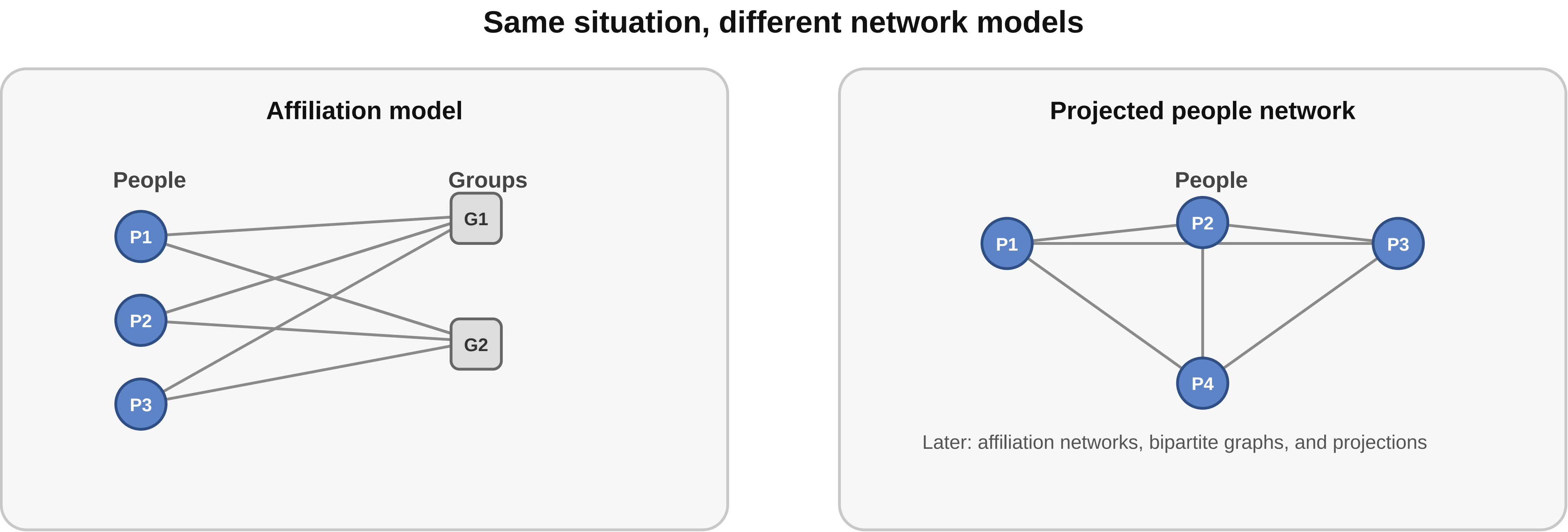
What Do All These Examples Have in Common?

- There are **things** (people, pages, institutions, cities).
- There are **connections** between them (friendships, links, loans, roads).
- The pattern of connections creates **structure** that matters.

$$\text{Network} = \text{Entities} + \text{Relationships between them}$$

What Gets Lost in a Network Model?

When we turn a real situation into a network, we always leave something out.



> The same relational setting can be modeled in more than one way. An affiliation structure can become a bipartite network or a projected one-mode graph depending on the question.

Speaker notes

Modeling is fundamentally an act of abstraction. When we translate a real-world situation into a network, we make choices about what to emphasize and what to ignore. A single social setting can be modeled as a bipartite affiliation network (people linked to groups) or as a projected one-mode network (people linked to each other via shared groups). Each modeling choice highlights different properties of the system.

Takeaway

> A network is a structured collection of entities and the relationships between them. Understanding networks means understanding how to manage, describe, and reason about these entities and their connections.

Speaker notes

Network science provides a universal language for describing relational systems. By focusing on the core abstraction—nodes and edges—we can move beyond domain-specific terminology to find patterns and principles that repeat across social, technical, economic, and biological systems.

Quick Check

Which of the following statements about networks are correct?

- A network consists of entities (nodes) and relationships between them (edges)
- The same network abstraction applies across very different domains
- Networks always have directed edges
- A network model captures every detail of the underlying system
- Edges can carry additional information such as weight or type

Speaker notes

Initial assessment of basic network definitions. It is important to distinguish between the entities (nodes) and their relationships (edges), and to recognize that models are simplified abstractions that necessarily trade off detail for analytical clarity.

Quick Check — Solution

Which of the following statements about networks are correct?

- **Correct:** A network consists of entities (nodes) and relationships between them (edges)
- **Correct:** The same network abstraction applies across very different domains
- **Incorrect:** Networks always have directed edges (edges can also be undirected)
- **Incorrect:** A network model captures every detail of the underlying system (they are models and have lossy abstractions)
- **Correct:** Edges can carry additional information such as weight or type

Impacts of Networks

> **Guiding Question:** Why do networks matter — what can we learn from connections?

The Whole Is More Than Its Parts

- Knowing every person in a city does not tell you how information travels — you need to know **who talks to whom**.
- Knowing every website does not tell you which ones are easy to find — you need to know **which pages link to which.**
- Knowing every bank does not tell you which failure would cause a cascade — you need to know **who has financial exposure to whom.**

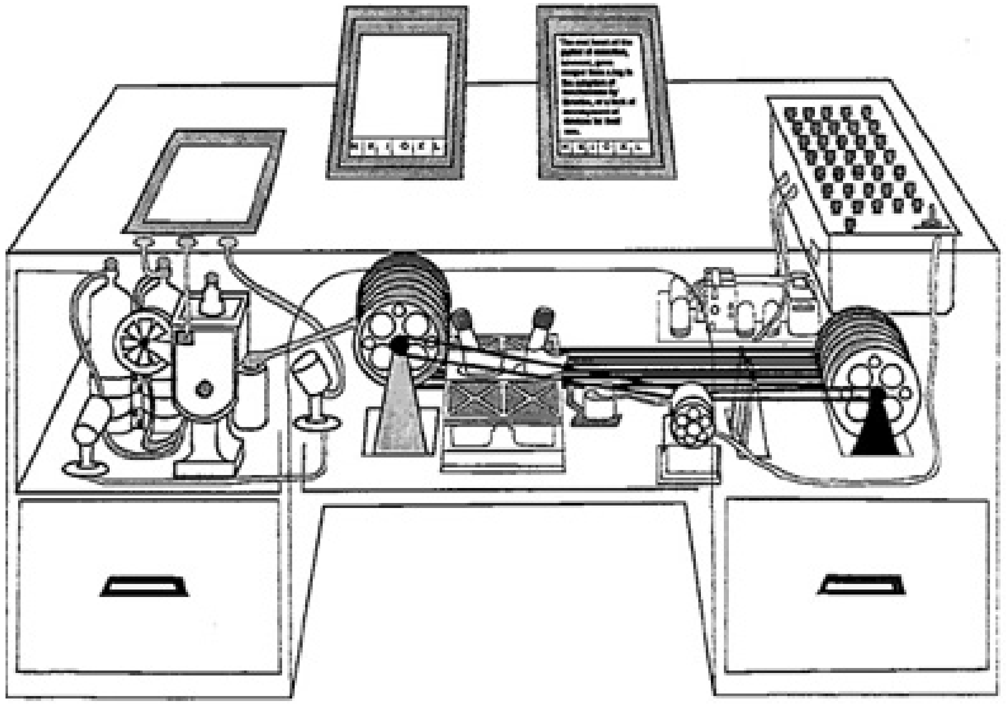
$$\text{Outcome} \neq \text{actor properties alone}$$

Speaker notes

The core premise of network science is that a system's behavior cannot be explained solely by the properties of its individuals. Whether we are analyzing social influence, web navigation, or financial stability, the structure of connections (the "topology") creates emergent outcomes that are invisible when looking at actors in isolation. This perspective shifts our focus from "what is this node?" to "how is this node connected?".

Practical Example: Linked Information

Vannevar Bush's MEMEX anticipated linked information systems and associative navigation (Bush, 1945).



Source: Bush 1945 — public-domain

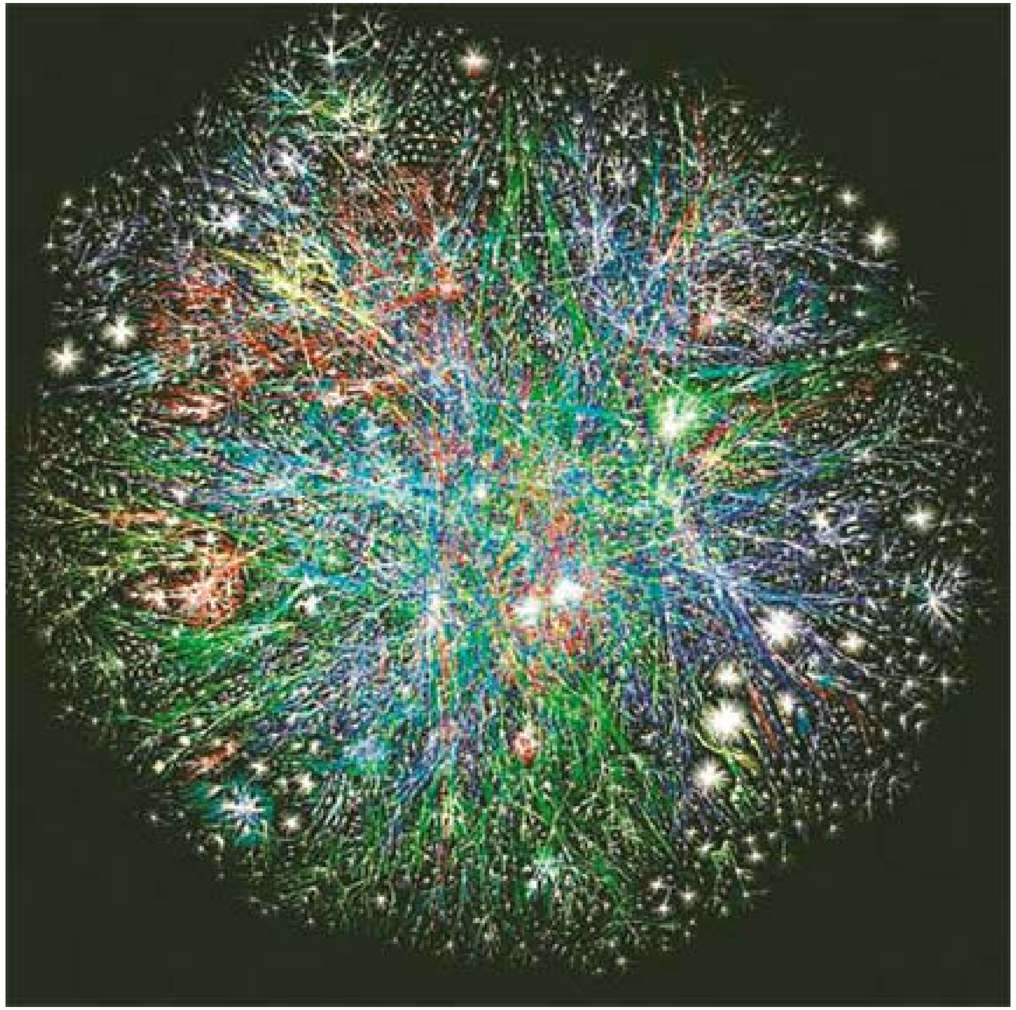
> **Vannevar Bush's MEMEX (1945).** A conceptual device for storing and linking documents through associative trails. It anticipated the idea that information becomes valuable through connections — the intellectual ancestor of hypertext and the Web.

Speaker notes

Vannevar Bush's MEMEX (1945) introduced the revolutionary idea of associative navigation. It shifted the focus from static document storage to dynamic, linked information pathways. This conceptual leap anticipated the development of hypertext and the World Wide Web, recognizing that the true value of information lies in its connectivity and the trails of association between nodes.

The Web as a Network

The Web operationalized hypertext at planetary scale: linked documents, user navigation, and a growing information network (Berners-Lee, 1990).



Source: Easley & Kleinberg 2010

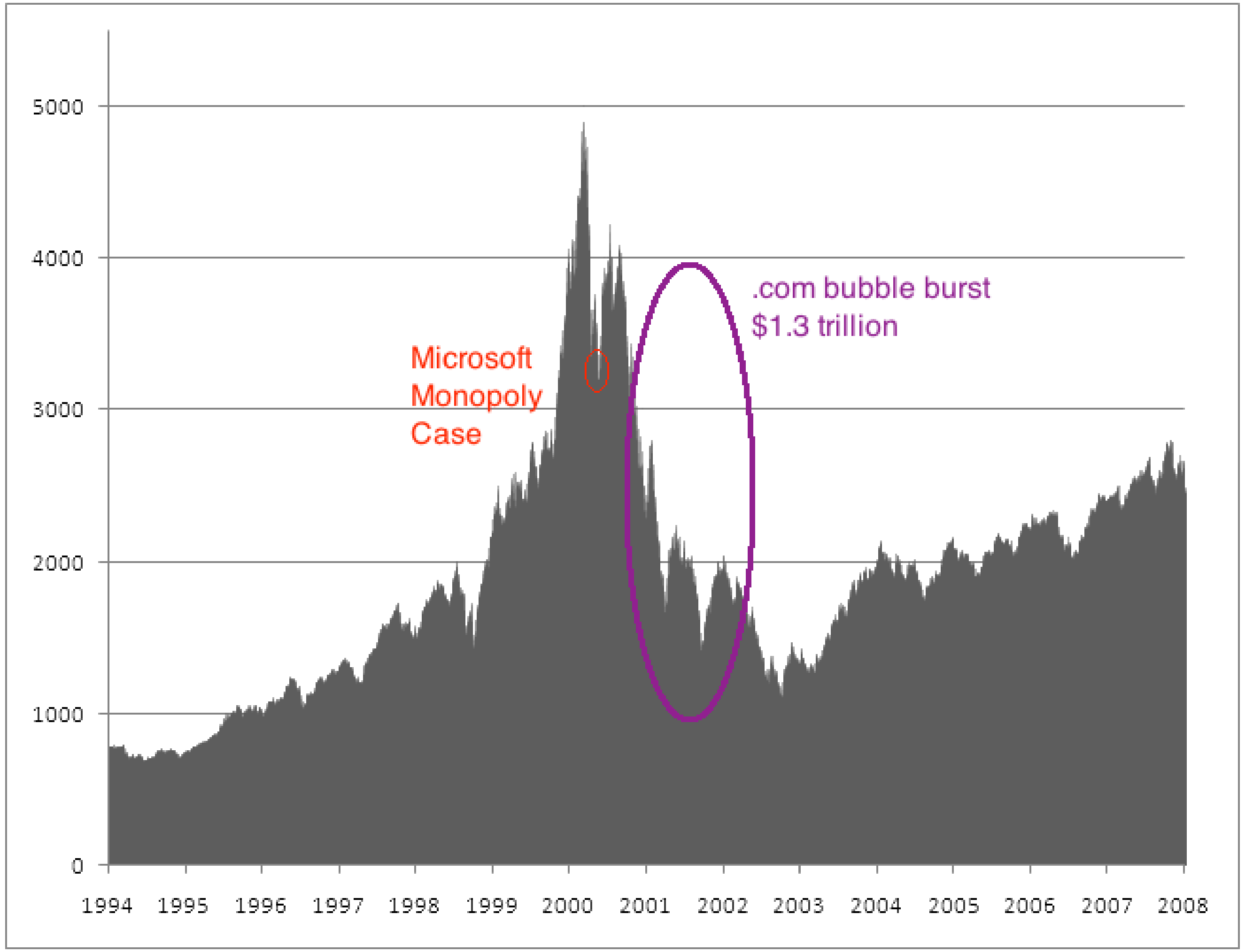
Visualization of internet routing paths. Each node represents a router or autonomous system, edges are data connections. The dense core and sparse periphery illustrate the uneven topology of real technical networks.

Speaker notes

The World Wide Web transformed the abstract idea of a network into a planetary-scale information infrastructure. By analyzing the Web's routing paths and hyperlink structures, we see a distinct uneven topology: a dense core of highly connected "major" nodes and a vast periphery of less accessible pages. This structural reality determines global visibility and attention flow.

Practical Example: Platform Power

Surviving web companies learned to exploit network effects — users became part of the value-creation process (O'Reilly, 2005).



Source: Public market data, NASDAQ composite index

NASDAQ composite index around the dot-com bubble (1995–2005).
The sharp rise and crash illustrate how many early web companies failed. The survivors — Google, Amazon, Facebook — were those that understood and exploited network effects.

Speaker notes

In the digital economy, "network effects" are a primary driver of value and platform survival. A system with network effects becomes more valuable as more users join, creating a "winner-take-all" dynamic. Companies like Google and Amazon succeeded precisely because their models exploited these relational dynamics to create strong lock-in and high barriers to entry.

What Can We Learn from Networks?

- **Who is important?** → centrality, influence
- **Are there groups?** → communities, clusters
- **How do things spread?** → diffusion, contagion
- **How robust is the system?** → fragility, resilience
- **What shapes attention?** → visibility, ranking, filtering

Speaker notes

Modern network analysis provides formal methods for answering critical structural questions. We can quantify importance (centrality), detect latent groups (communities), model spreading processes (diffusion), and assess a system's vulnerability to failure (robustness). Each of these analytical lenses corresponds to a specific dimension of network structure.

Takeaway

> Networks matter because structure shapes visibility, coordination, influence, and lock-in. Looking at connections reveals what looking at individuals alone cannot.

Speaker notes

A network is not merely a visualization of relationships; it is a structural skeleton that constrains and enables actor behavior. By studying these connections, we gain analytical power to predict outcomes that remain hidden when focusing exclusively on individual components.

Quick Check

Why are network models useful for studying platforms like the Web?

A. They allow us to ignore structure and focus on individual users.

B. They reveal how connections create effects like lock-in and influence.

C. They primarily help in analyzing the content of individual web pages.

D. They show that all users have equal influence in a system.

Speaker notes

Review of the core value of network modeling. The focus remains on how links—rather than just node content—determine the functional characteristics and power dynamics of large-scale systems like the Web.

Quick Check — Solution

Why are network models useful for studying platforms like the Web?

Correct: B. They reveal how connections create effects like lock-in and influence.

The Web's power comes not just from the individual pages (nodes), but from how they are linked (edges), which drives visibility and platform power.

Network Science & Network Analysis

> **Guiding Question:** What phenomena can we actually analyse with networks?

From Intuition to Analysis

We already said networks reveal structure. But what kinds of questions can we actually answer?

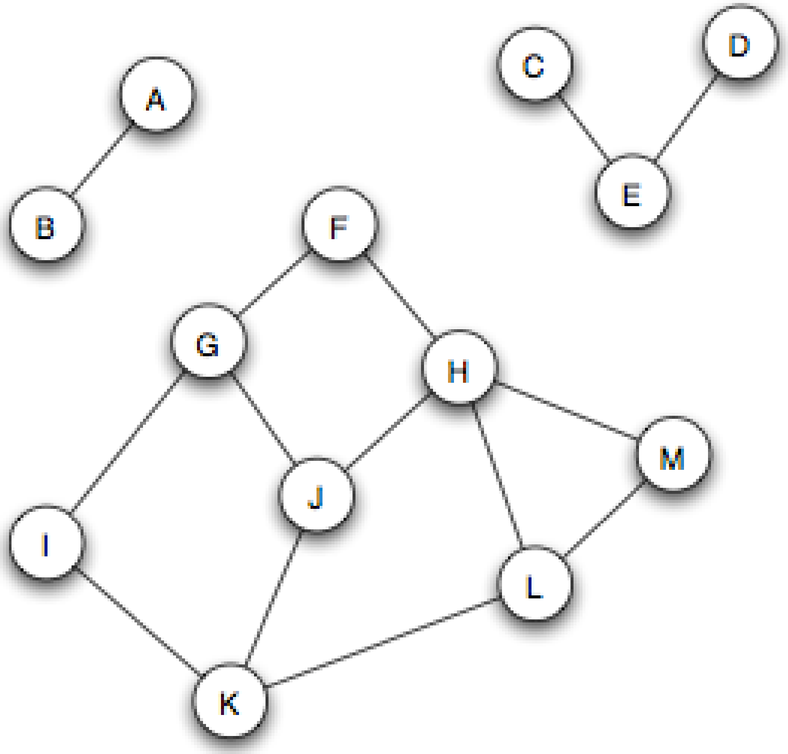
- **Structure**: How is the network organized? Are there clusters, hubs, or bottlenecks?
- **Position**: Who sits in a central or peripheral position? Who bridges groups?
- **Dynamics**: How does information, disease, or opinion spread through the network?
- **Evolution**: How does the network change over time? What drives growth?

Speaker notes

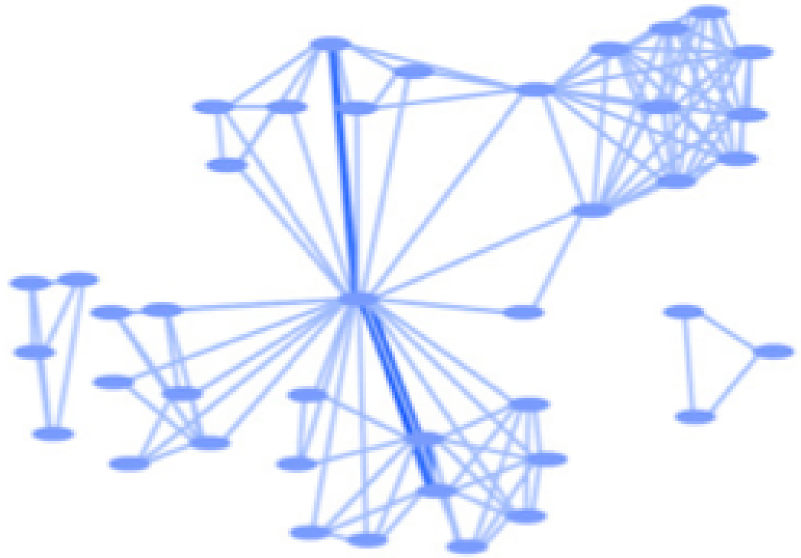
Transitioning from intuition to formal analysis requires categorizing structural questions into four main pillars: Structure (the global organization), Position (the role of individuals), Dynamics (how things move through the network), and Evolution (how the network grows). Each pillar is supported by specific mathematical and algorithmic frameworks.

Analysing Structure: Communities

Networks often have dense internal clusters that correspond to real groups.



Source: Easley & Kleinberg 2010



Source: Easley & Kleinberg 2010

Three separate connected components — groups of nodes with no links between them.

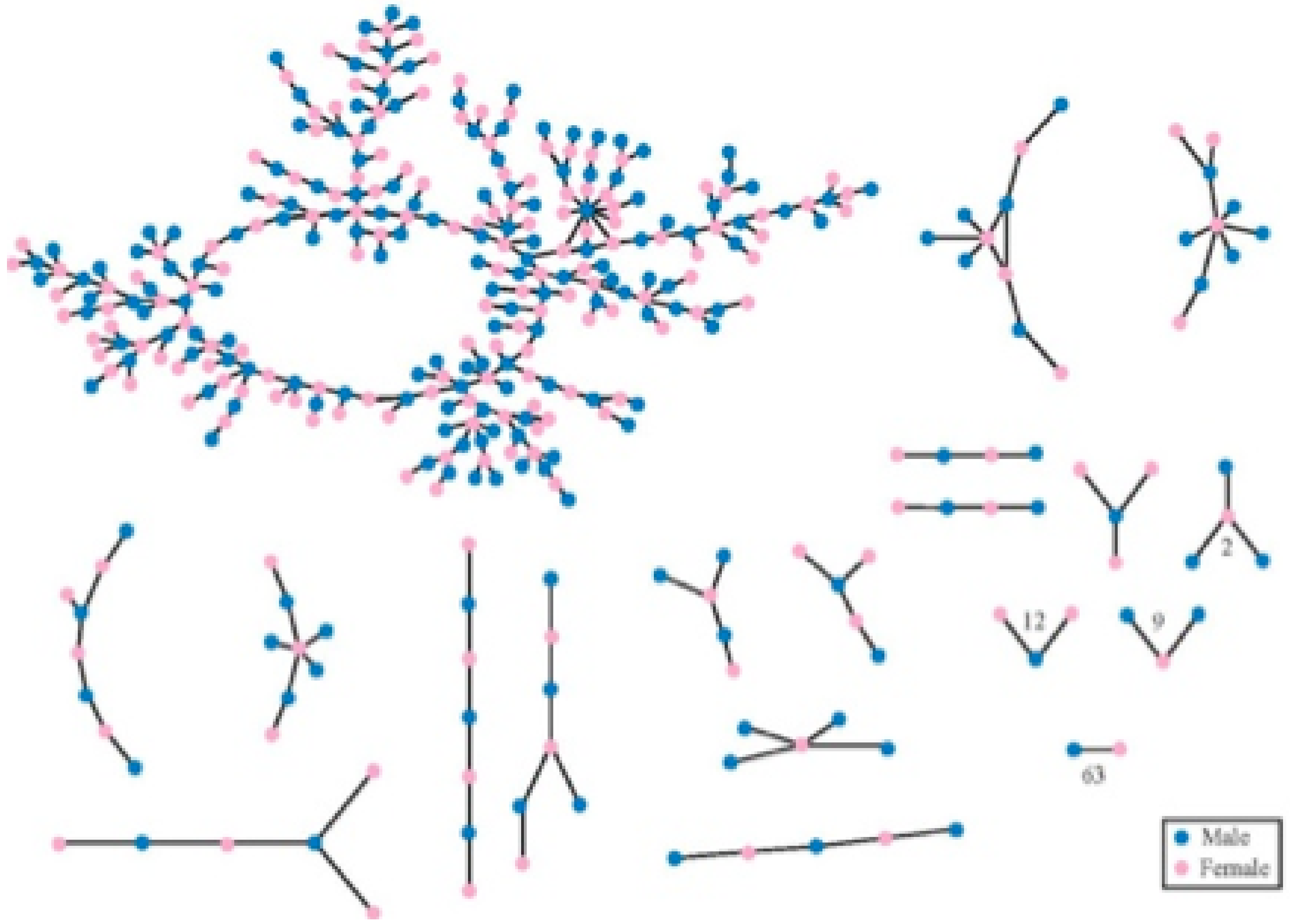
A scientific collaboration network where clusters represent research groups.

Speaker notes

Structural analysis often centers on "communities"—subgroups where nodes are more densely connected to each other than to the rest of the network. Detecting these communities allows us to identify functional units, such as research labs in a collaboration graph or social cliques in a high school, revealing the latent organization of the system.

Analysing Dynamics: Diffusion and Spreading

How a rumor, virus, or innovation spreads depends critically on network structure.



Source: Easley & Kleinberg 2010

High school friendship network.
Nodes are students, edges are reported friendships. The structure shows a dense core and peripheral chains — a rumor starting in the core spreads far faster than one starting at the edge.

Speaker notes

The dynamics of a network describe how processes like rumors, viruses, or innovations move across edges. The speed and extent of this spread are not uniform; they are heavily influenced by the network's topology. A rumor in a dense friendship core spreads differently than one in a sparse periphery, motivating the need for diffusion modeling.

Core Questions of Network Science

Across all these domains, network science asks a common set of structural questions:

- How do we **describe** a network formally?
- Which structural **properties** matter?
- How do networks shape **diffusion** and coordination?
- Which common **patterns** appear across different domains?

Speaker notes

The core objective of this course is to develop a robust vocabulary and toolkit for answering these structural questions across different domains. By focusing on universal patterns—like scale-free distributions or small-world effects—we can apply insights from one discipline to solve complex relational problems in another.

Takeaway

> Network science gives us concrete tools to analyse structure, position, dynamics, and evolution. The same analytical questions apply whether we study friendships, web links, financial flows, or disease transmission.

Speaker notes

Network science is a deeply applied field. The tools you will learn are used by data scientists, epidemiologists, and economists to solve real-world problems. Whether the goal is to optimize a transport network or to stop a financial contagion, the analytical foundations remain the same.

Quick Check

Match the network phenomenon with the type of analysis used to study it:

1. Finding groups of friends in a high school
2. Measuring how fast a rumor spreads
3. Identifying the most influential blogger

A. Dynamics (Diffusion) B. Structure (Communities) C. Position (Centrality)

Speaker notes

Final module assessment matching real-world phenomena to their corresponding analytical concepts. This reinforces the distinction between studying groups (structure), spreading (dynamics), and individual importance (position).

Quick Check — Solution

Match the network phenomenon with the type of analysis used to study it:

- **1 - B. Structure (Communities)**
- **2 - A. Dynamics (Diffusion)**
- **3 - C. Position (Centrality)**

Curriculum

> **Guiding Question:** What do you want to learn about networks?

What You Will Learn

Course Topics

1. **Graph theory** — the formal language for describing networks
2. **Social networks** — ties, communities, and information flow
3. **Centrality and influence** — who matters and why
4. **Community detection** — finding groups in networks
5. **Information networks and the Web** — links, search, and ranking
6. **Network dynamics** — diffusion, contagion, and cascades

Speaker notes

The curriculum follows a logical progression from the formal language of graph theory to advanced topics in social and information networks. This structured approach ensures that you build a solid foundation before tackling complex dynamic processes like cascades and diffusion.

Course Goal

Use network analysis to understand and design social information systems.

- not just learn definitions — **interpret** formal concepts
- not just run algorithms — **critique** the models behind them
- not just describe networks — **reason** about their consequences

Speaker notes

The ultimate goal is applied conceptual fluency. You should be able to explain not only the mathematical definitions but also their broader implications for designing and critiquing modern social information systems. Look for the "why" behind every "what".

Recommended Reading

- David Easley and Jon Kleinberg, *Networks, Crowds, and Markets* (Easley & Kleinberg, 2010)
- Mark Newman, *Networks: An Introduction* (Newman, 2010)

Speaker notes

These two textbooks offer complementary perspectives: Easley & Kleinberg focus on high-level sociological and economic intuitions, while Newman provides a more rigorous, systematic foundation in network mathematics. Use both to build a well-rounded understanding.

How to Learn This Course

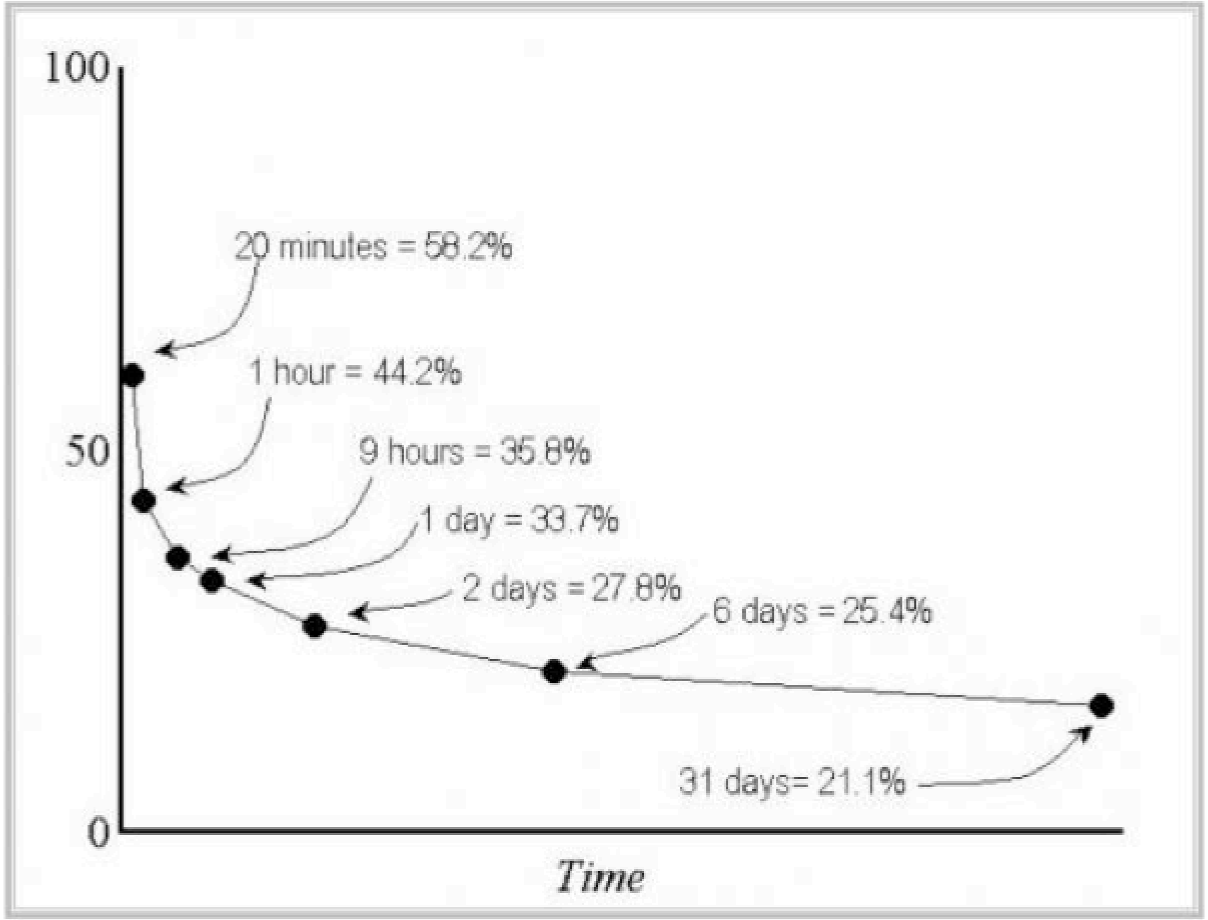
- **preview** slides before class
- **discuss** key ideas in small groups
- **work through** exercises yourself before comparing solutions
- **use AI tools** to accelerate access to explanations — but if you cannot explain a concept without the tool, you do not yet own it

Speaker notes

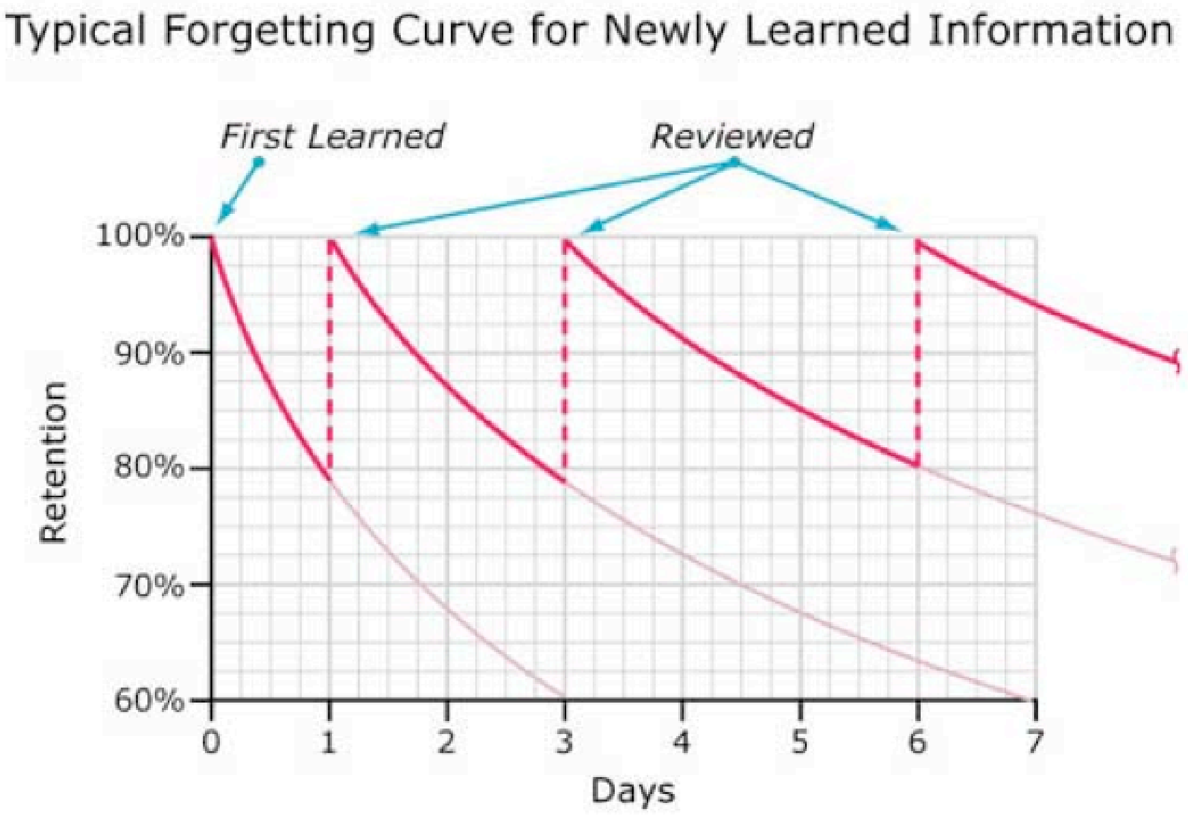
Effective learning in this course depends on active engagement. Use the slides for previewing, but focus on the exercises and group discussions to internalize the material. While AI tools can aid in finding quick definitions, true ownership of the concepts requires the ability to explain and apply them independently.

Forgetting and Repetition

Repeated contact with the same concept reduces forgetting and improves retention.



Source: Ebbinghaus forgetting curve, redrawn



Source: Ebbinghaus curve with spaced repetition, redrawn

Speaker notes

Retention of complex relational concepts is improved through spaced repetition. The course structure and exercises provide multiple opportunities to revisit and stabilize key ideas over time.

Takeaway

> The course is about modeling, interpretation, and critique — not only about definitions. You will build a toolkit for understanding networks that applies across social, technical, and economic systems.

Speaker notes

In summary, this course is designed to transition you from intuitive graph thinking to formal, critical network analysis. You are building a toolkit that will remain relevant across any relational system you encounter in your professional or academic career.

Image Credits & References

Adamic, Lada A. and Glance, Natalie (2005). The political blogosphere and the 2004 U.S. election: divided they blog. *Proceedings of the 3rd International Workshop on Link Discovery (LinkKDD '05)*.

Bush, Vannevar (1945). As We May Think. *The Atlantic Monthly*.

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning About a Highly Connected World. *Cambridge University Press*. https://www.cs.cornell.edu/home/kleinber/networks-book/

Zachary, Wayne W. (1977). An Information Flow Model for Conflict and Fission in Small Groups. *Journal of Anthropological Research*.